

Task 4 - Sentiment Analysis Report

Objective

The objective of this project is to develop a Sentiment Analysis model capable of classifying Twitter posts into Positive, Neutral, and Negative sentiments. The project utilizes Natural Language Processing (NLP) techniques and Machine Learning algorithms to analyze textual data and extract meaningful insights.

Dataset

Dataset Used: Twitter_Data.csv

The dataset contains Twitter posts along with sentiment labels:

- Positive (1)
- Neutral (0)
- Negative (-1)

Tools and Technologies

- Python
- Pandas
- NumPy
- Matplotlib
- Seaborn
- NLTK
- WordCloud
- Scikit-Learn

Methodology

1. Data Understanding

- Loaded and explored the dataset.
- Analyzed dataset structure, missing values, and sentiment distribution.

2. Data Cleaning

- Removed missing values.
- Removed duplicate records.
- Converted sentiment labels into readable categories.

3. Text Preprocessing

- Converted text to lowercase.

- Removed URLs.
- Removed numbers and special characters.
- Removed stopwords.
- Removed extra spaces.

4. Exploratory Data Analysis

- Sentiment Distribution Analysis.
- Tweet Length Distribution.
- Average Tweet Length by Sentiment.
- Positive, Negative, and Neutral Word Clouds.

5. Feature Engineering

- Applied TF-IDF Vectorization to transform text into numerical features.

6. Machine Learning Models

The following models were implemented:

- Naive Bayes
- Logistic Regression
- Support Vector Machine (SVM)

7. Model Evaluation

Models were evaluated using:

- Accuracy Score
- Classification Report
- Confusion Matrix

Results

The sentiment classification model achieved approximately 86% accuracy on the test dataset. Among the evaluated models, the best-performing model was selected for final sentiment prediction.

Conclusion

This project successfully demonstrates the application of Natural Language Processing and Machine Learning techniques for sentiment classification. The model effectively predicts whether a tweet expresses a positive, neutral, or negative sentiment and provides valuable insights into public opinion and social media trends.